Midterm 2: Study Guide & Plan

Stats 280, Lectures 9–15 (Chapters 13–17) plus R for probability and simulation. Exam: Wednesday, November 11.

How the real exam is built (W25 and W26)

Topic	W25	W26	What they ask
Sequential draws without replacement	Q1 cards until a spade	Q1 The Traitors	multiplication rule, complements, "more than k", case-splitting
Bounds, independence, Venn diagrams	Q2	Q2	max/min of P(A and B), P(A or B), the independence check, 3-set Venn counts
Bayes / law of total probability	Q3 butterflies	Q3 study habits	tree diagram, P(category | evidence) for each category
Hypergeometric vs binomial + R	Q4	Q4	choose formulas, `dhyper/phyper/dbinom/pbinom`, the off-by-one rules
Random variables + R simulation	Q5 cointoss, draw1	Q5 simM	distribution table, E, SE, write a function, fill blanks, "what does this estimate"
Box model + normal approximation	Q6 Celtics	Q6 red balls	EV, SE, continuity correction, exact R command, interpretation
Extra credit proof	Q7 E(X+Y)	—	derive a property from definitions

Formulas to know cold

- $P(A|B) = P(A \text{ and } B)/P(B)$; multiplication rule $P(A \text{ and } B) = P(A)P(B|A)$; addition rule $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.
- Bounds: $\max(P(A)+P(B)-1, 0) \le P(A \text{ and } B) \le \min(P(A), P(B))$ and $\max(P(A), P(B)) \le P(A \text{ or } B) \le \min(P(A)+P(B), 1)$. For three events, the minimum of the triple overlap is $P(A)+P(B)+P(C)-2$ (if that's positive).
- Three-set inclusion–exclusion: $P(A \cup B \cup C) = \sum P(\cdot) - \sum P(\text{pairs}) + P(A \cap B \cap C)$.
- Total probability $P(B) = \sum_i P(B|A_i)P(A_i)$; Bayes $P(A|B) = P(B|A)P(A)/P(B)$; posterior odds = prior odds × data odds; $p = o/(1+o)$.
- Counting: $n!$, permutations $n!/(n-k)!$, combinations $\binom{n}{k}$. Arranging letters with repeats: choose positions step by step.
- Binomial $\binom{n}{k}p^k(1-p)^{n-k}$; hypergeometric $\binom{m}{x}\binom{n}{k-x}/\binom{m+n}{k}$.
- $E(X) = \sum x\,P(X=x)$; $\text{Var}(X) = E(X^2) - E(X)^2$; $\text{Cov}(X,Y) = E(XY) - E(X)E(Y)$; $\text{Var}(X+Y) = \text{Var}X + \text{Var}Y + 2\text{Cov}$; $SE(cX) = |c|SE(X)$.
- $E(X+Y) = E(X) + E(Y)$ **always**. Variances add only when the covariance is 0 (e.g. independent).

- Box models (draws with replacement): sum EV $= n\mu$, SE $= \sqrt{n}\,\sigma$; average EV $= \mu$, SE $= \sigma/\sqrt{n}$; 0-1 box SD $= \sqrt{p(1-p)}$; two-value box SD $= (b-s)\sqrt{p_b p_s}$.
- Without replacement: EV of the sum is still $n\mu$, but the SE formula does **not** apply. Simulate instead.
- Normal approximation for a count: use EV and SE, apply the continuity correction (±0.5), and read area within ±z from the table. Rule of thumb: $np > 5$ and $n(1-p) > 5$.

Traps that cost points

- Independent ≠ mutually exclusive. Mutually exclusive events with positive probabilities are always *dependent*.
- "Second card is an ace" has unconditional chance 4/52, by symmetry. Don't condition on things you weren't told.
- "At least one" → 1 − P(none). "More than 5 draws until the first X" → the first 5 draws contain no X.
- Without replacement → hypergeometric (`dhyper`); with replacement or independent trials → binomial (`dbinom`).
- `pbinom(k)` includes k. "At least 60" is `1 - pbinom(59, ...)`, not 60.
- `dhyper(x, m, n, k)`: the third argument is the number of *failure* tickets, not the box size.
- Continuity correction direction: "at least 330" → 329.5; "at most 70" → 70.5; "between 50 and 70" → 49.5 to 70.5.
- For a conditional probability by simulation, filter first: `mean(A[B])`, not `mean(A & B)`.
- SE from simulations: `sd(x) * sqrt((N-1)/N)`. `return(x)` needs parentheses. `sample()` defaults to *no* replacement.
- Law of averages: the absolute chance error grows like $\sqrt{n}$; the percentage chance error shrinks like $1/\sqrt{n}$.

Plan: Sunday night to Wednesday

When	Do
Sunday night	R Bootcamp Modules 0–4. Every drill on paper first, then check in R.
Monday	R Bootcamp Modules 5–8: Gauntlet timed, flashcards to 40/40. Then the staff's **W26 Midterm 2**, timed (80 min). Grade it with the official key.
Monday night	**Practice Midterm 2A**, timed. Grade it, and write down every point lost and why.
Tuesday	**2B** in the morning and **2C** in the evening, both timed. Redo every miss from scratch. One flashcard round before bed.
Wednesday	Read this sheet plus the Bootcamp key's one-page summary. One flashcard round. Nothing new.

Pace: about 24 seconds per point, so roughly 12 minutes per 30-point question and 16 per 40-point one. Do the R parts first in each question: they're fast, and they need no justification.